

Java & Spring

A short demo book for Readora's virtual-edition reader.

This is placeholder content generated for demonstration purposes only, showing the in-app PDF viewer end to end.

Chapter 1: Why Java

Java is a statically typed, object-oriented language that runs on the JVM. Its "write once, run anywhere" promise made it a staple of enterprise backend systems.

Key traits: strong typing, garbage collection, a huge standard library, and a mature ecosystem of tooling.

Classes and Objects

A class is a blueprint; an object is an instance of it.

```
public class Book {  
    private String title;  
    public Book(String title) {  
        this.title = title;  
    }  
}
```

Chapter 2: The JVM

Java source compiles to bytecode, which the JVM then interprets or JIT-compiles to native machine code.

This is what gives Java its portability: the same .class file runs unmodified on any platform with a JVM.

Garbage Collection

The JVM automatically reclaims memory for objects that are no longer reachable from any live reference.

Modern collectors (G1, ZGC) aim for short, predictable pause times even on large heaps.

Chapter 3: Why Spring

Spring is a framework built around dependency injection and inversion of control: objects declare what they need, and the container wires them together.

This makes large codebases easier to test and compose.

Spring Boot

Spring Boot layers convention-over-configuration on top of Spring: sensible defaults, an embedded server, and starter dependencies that pull in what you need.

```
@SpringBootApplication
public class App {
    public static void main(String[] a) {
        SpringApplication.run(App.class, a);
    }
}
```

REST with Spring MVC

`@RestController` classes expose HTTP endpoints. Spring handles request routing, JSON (de)serialization, and validation, letting you focus on business logic.

```
@GetMapping("/books/{id}")  
public Book get(@PathVariable UUID id) { ... }
```


Data with Spring Data JPA

Spring Data JPA generates repository implementations from interface method names, turning common queries into a single line of code.

```
interface BookRepository  
    extends JpaRepository<Book, UUID> {}
```

Closing Notes

This concludes the Java & Spring demo edition.

It exists to exercise Readora's virtual-edition purchase and in-app reading flow end to end, not as real learning material.

Thanks for reading.